Smart Fall Prevention in Domestic Environments

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***Abstract*—Falls represent a critical challenge in global healthcare, frequently leading to severe injury and mortality among older populations, many of whom never fully recover. For elderly individuals, roughly 28 to 35 percent of adults over the age of 65 fall at least once a year, and that rate only grows with age. For people who are visually impaired, the danger compounds in a different way: they cannot see what they are about to step on. The hazards are rarely dramatic. They could present themselves as a cord left across a doorway, a carpet edge that has curled up over time, or a bag someone forgot on the floor. These are the things that actually cause falls indoors, and assistive technology has largely overlooked them. Current solutions are generally divided into two categories: wearable sensors that confirm a fall has occurred, and outdoor navigation systems that help avoid objects like parked cars. Much of the current solution set focuses on fall detection but rarely addresses the fall prevention problem. Previously, this problem remained untackled due to significant technical challenges in designing a portable, reliable detection device for a constantly changing environment. Computer vision systems have struggled to detect small, low-contrast hazards at varying depths, such as wires, carpets, and stairs, especially in real-time scenarios without expensive, bulky hardware. Due to recent advances in edge AI systems enabling real-time on device detection, we present a fall-prevention cane for detecting wire hazards in indoor environments for the elderly and visually impaired.**

I. Introduction

Falls represent one of the most pressing public health challenges, especially as the elderly population continues to grow. Although falls occur across all populations, they are particularly prevalent in ages over 65, with approximately 30% experiencing at least one fall per year [3]. These falls can lead to debilitating injuries, hospitalization, and even death, all of which drive up medical costs and stress. While falls can happen across all environments, 55% of fall-related injuries occur within the home [2]. Falls arise from a variety of behavioral and environmental issues, such as cognitive decline or hazards like strewn wires.

Current assistive technologies have focused on detecting falls and prompting requests for help to ensure an adequate level of care rather than preventing falls. Environmental risk factors, such as loose objects, poorly placed furniture, and misplaced items, have been identified as contributors to falls [2]. But real-time detection and measurement still haven't been fully explored due to the technical challenges. These challenges include the resource constraints of portable edge devices, such as limited compute and the need for continuous real-time inference. Recent advances in edge devices have made this possible, enabling us to present a fall-prevention smart cane that focuses on real-time detection of wires in indoor environments.

II. Related Work

Much of the previous work in mobile health and falls focuses on fall detection, but there is growing interest in fall prevention. These existing systems aim to confirm that a fall has occurred and prompt for emergency response, ensuring proper continuation of care. These approaches often use wearables, such as IMUs, to detect falls, or unique wearables, such as skin sensors or sole inserts, to track gait [4,5]. Another approach is integrated camera systems within the home, pressure systems, and Wi-Fi-based motion tracking [4,5]. While both approaches are effective at fall detection and proper response, they still do not address the underlying problems that cause falls, such as hazards.

Previous studies on fall prevention have chiefly focused on fall risk screening, particularly on physiological factors such as balance, gait, and cognitive function [1]. Although environmental risk factors have been identified, many fall risk apps and mobile health solutions have failed to objectively measure them [1]. Other work in environmental fall prevention has focused on identifying home features that contribute to fall risk and how home modification and assessments reduce these risk factors [2]. These high-risk environmental hazards are often slippery surfaces, worn carpets, staircases, loose objects, and poorly designed furniture [2].

Managing environmental risk factors is prevalent but mostly static and preventive, rather than focusing on real-time detection, as home conditions are dynamic and can change over time. Camera-based systems have shown promise in bridging this gap. Adjacent fields, such as smart canes for the visually impaired, have shown great promise for camera-based object detection and depth estimation, providing real-time obstacle warnings [4]. While fixed camera systems have been shown to help, these computer vision systems often involve constant surveillance through in-room cameras. Together, these works highlight a gap in environmental hazard detection for fall prevention and the feasibility of a low-cost dynamic solution for real-time prevention.

III. System Design

The primary goal of this system is practical deployment: it must be low-cost, lightweight, and simple enough for elderly and visually impaired users to carry as part of a standard walking cane. The full system consists of three subsystems: a sensing and compute unit mounted on the cane, a detection model that runs on demand, and an alert mechanism delivered through a local web interface. Fig. 1 shows the physical cane concept, and Fig. 2 illustrates the system-level architecture.

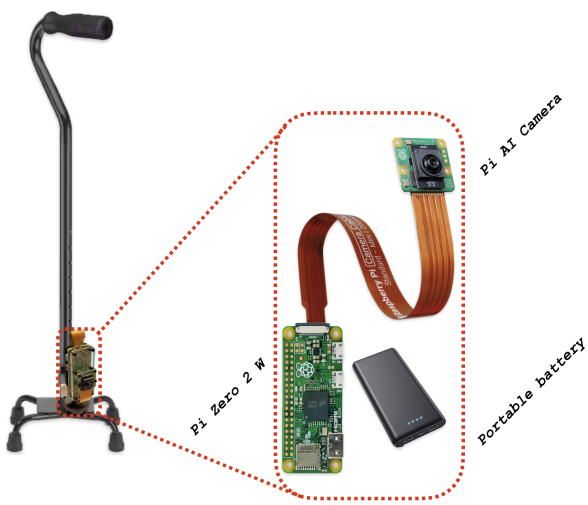

Fig. 1. Fall Prevention Cane Concept Design

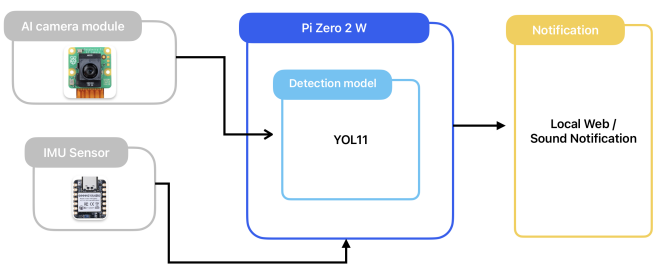

Fig. 2. Fall Prevention Cane System Design

A. Hardware Platform

The computing platform is a Raspberry Pi Zero 2 W, selected for its small form factor, low weight, built-in Wi-Fi, and low cost ($15). While less powerful than other single-board computers it is sufficient to coordinate sensor input, trigger inference, and serve detection results and be embedded in a cane without adding meaningful bulk.

Visual sensing is handled by a Raspberry Pi AI Camera with a Sony IMX500 image sensor, performing on sensor inference integrated neural processing unit. The camera connects to the Pi Zero 2 W via a CSI ribbon cable.

Because the Pi Zero 2 W does not include an onboard inertial measurement unit (IMU), an external LSM6DS3TR-C sensor is added to the board. The LSM6DS3TR-C provides 3-axis accelerometer and 3-axis gyroscope data over I2C and is responsible for detecting the cane's motion state.

The hardware assembly is mounted at the lower portion of the cane, with the camera angled forward and positioned 2 ft above the ground. This field of view was chosen to capture the area directly ahead of the user's next few steps, where ground-level hazards such as electrical cords and raised carpet edges are most likely to appear.

B. IMU-Triggered Inference

A naive approach to hazard detection would have the model continuously firing alerts for every hazard detected on every incoming frame. To remove this redundancy and ensure we send alerts at the correct time, we utilize the cane's active state with an IMU. When the cane is in motion, either swinging forward or being lifted, the camera's field of view shifts rapidly, and detections are less reliable, meaning continuous alerting would be excessive noise.

Instead, the system uses the LSM6DS3TR-C IMU module from the XIAO Sense to determine when the cane is in a stable, grounded state, and only then triggers an alert delivery. The logic proceeds in two stages.

First, the system monitors the accelerometer's Z-axis to determine whether the cane is at a consistent height, which indicates that the tip is resting on the ground rather than being lifted or swung. Second, once the cane is grounded, the system evaluates the magnitude of acceleration across all axes to ensure it is not in motion before an alert is issued.

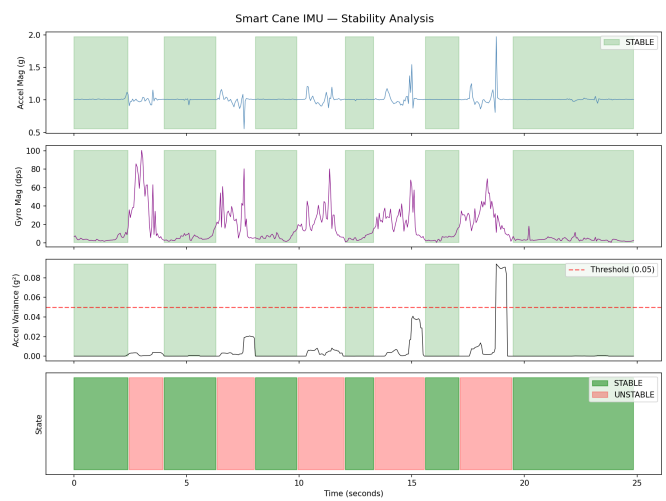

Fig. 3. IMU data collection. The four charts represent the magnitude of acceleration, gyroscope, acceleration variance, and stability classification in descending order.

C. Detection Model

The detection model is YOLO11n (nano), selected for its strong balance between accuracy and model size. YOLO11n is the smallest variant in the YOLO11 family and is designed for deployment on edge devices with limited compute budgets.

The model was trained using Roboflow on a custom dataset targeting a single class: wires (power, USB, extension). This narrow scope improves reliability given limited data. The dataset included 114 training images, 11 validation images, and 5 test images, with augmentation (brightness, blur, flipping) to expand variability.

The trained model was exported to a format compatible with the IMX500's neural processing unit. At inference time, the Pi Zero 2 W sends a capture command to the AI Camera, which acquires a frame and runs YOLO11n entirely on-sensor. The camera returns bounding box coordinates and confidence scores to the Pi, which then evaluates whether a hazard is present based on a confidence threshold.

D. Alert Delivery

When the detection model identifies a hazard above the confidence threshold, the system delivers an audio alert to the user. The Pi Zero 2 W runs a local web server that generates the alert sound directly. Upon a positive detection, the server triggers an audio cue that warns the user of a hazard in the path ahead.

This web-based architecture was chosen not only for ease of implementation but also for extensibility. Because the alert logic runs through a networked interface, the system can be extended in future iterations to push notifications to a guardian's device or a remote monitoring dashboard, enabling caregivers to be aware of detected hazards or repeated near-miss events. An example alert output is shown in Fig 4.

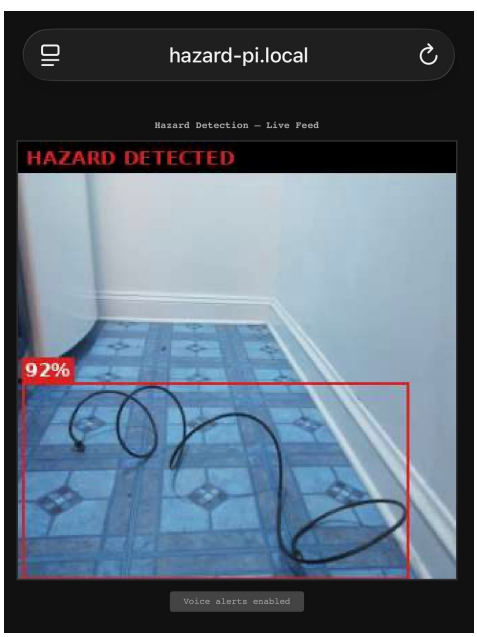

Fig. 4. Alert Delivery. The above alert shows a 92% confidence score for detecting a wire using our trained YOLO11n model.

IV. Experiment Setup

Our fall prevention system was evaluated along three axes: static detection across controlled environments, distance sensitivity, and a preliminary walking trial.

A. Static Detection

The cane and mounted camera system were fixed on a tripod with the hazard placed at a consistent distance of 60 cm. We tested two hazard types across four floor and lighting combinations, as well as no-hazard false positive checks, for 12 conditions total. Each condition was run 5 times. Results are shown in Table I.

TABLE I
Static Detection Trial Results (5 reps per condition, hazard at 60 cm)

Trial	Hazard	Floor	Lighting	Det.	Conf.
1a	USB	Wood	Bright	4/5	0.78
1b	USB	Wood	Dark	2/5	0.71
1c	USB	Carpet	Bright	4/5	0.81
1d	USB	Carpet	Dark	3/5	0.73
1e	USB	Tile	Bright	5/5	0.85
1f	USB	Tile	Dark	3/5	0.74
2a	Ext.	Wood	Bright	5/5	0.84
2b	Ext.	Wood	Dark	3/5	0.74
2c	Ext.	Carpet	Bright	5/5	0.88
2d	Ext.	Carpet	Dark	4/5	0.77
2e	Ext.	Tile	Bright	5/5	0.91
2f	Ext.	Tile	Dark	4/5	0.79
3a	FP	Wood	Bright	2/5	—
3b	FP	Wood	Dark	1/5	—
3c	FP	Carpet	Bright	0/5	—
3d	FP	Carpet	Dark	0/5	—
3e	FP	Tile	Bright	2/5	—
3f	FP	Tile	Dark	1/5	—

B. Distance Sensitivity

To characterize the effective warning range of the system, we tested detection rate at distances of 1-6 ft using an extension cord on a wood floor under bright lighting, with 5 repetitions per distance. Results are shown in Table II.

TABLE II
Distance Sensitivity — extension cord, wood floor, bright lighting (5 reps each)

Distance (ft)	Det. Rate	Mean Conf.	Mean Latency (ms)
1	2/5	0.72	244
2	3/5	0.76	245
3	5/5	0.86	247
4	5/5	0.89	248
5	5/5	0.87	246
6	3/5	0.74	245

C. Preliminary Walking Trial

To test real-world feasibility, three participants each walked a 6 m straight path three times with one USB cable and one extension cord planted at unknown positions. Alert mode was done through audio. Per walk, we recorded (1) whether the alert fired before the next step. After completing all three walks, the participant answered each question. Q1 - I had adequate time to react to the alert (1–5 scale) Q2 - The audio

alert was disruptive (1–5 scale) Q3 - The alerts were clear and easy to understand (1–5 scale) Results are shown in Table III.

TABLE III
PRELIMINARY WALKING TRIAL RESULTS (5 PARTICIPANTS, 3 WALKS EACH)

	Alert Before Contact			Survey (1–5)		
P	**W1**	**W2**	**W3**	**Q1**	**Q2**	**Q3**
P1	N	Y	Y	3	4	3
P2	Y	N	Y	3	5	4
P3	Y	Y	N	4	4	3
P4	Y	Y	Y	4	4	3
P5	Y	N	Y	3	5	4
Avg	4/5	3/5	4/5	3.4	4.4	3.4

Tested by Rama Janco, Sungwoon Kim, Andre Washington, Ethan Fan, Daniel Terrero

V. RESULTS

A. Static Detection

Tile floors achieved the highest detection rates and confidence scores, with extension cords detected 5/5 at 0.91 mean confidence under bright light, likely due to their uniform, high-contrast surface. Carpet performed similarly in bright conditions and remained stable in low light, as its texture lacks strong linear features. Wood floors performed worst, especially for USB cables in low light (2/5, 0.71 mean confidence), where grain patterns and low contrast hinder detection.

False positives followed the same pattern. Tile and wood each produced 2/5 false detections in bright conditions, caused by misclassification of straight edges such as wood and grout lines or appliance seams. Carpet produced no false positives, and dark conditions reduced false triggers across all surfaces due to lower contrast.

Observation: During static detection trials, white extension cords exhibited noticeably higher confidence scores in dark lighting conditions compared to darker-colored cables, suggesting that cable color is an independent factor in detection reliability, likely due to higher luminance contrast between a white cable and a dark floor under low ambient light. This finding was incidental and not part of the original experimental design; controlled replication across cable colors is recommended in future work.

B. Distance Sensitivity

Detection performance varied non-linearly with distance. At 1 ft, the wire occupied too much of the frame and lacked context, yielding 2/5 detections at 0.72 mean confidence. Performance improved at 2 ft and peaked between 3–5 ft, achieving 5/5 detections with 0.86–0.89 confidence—an optimal range where the wire is clear but still contextualized by the floor. At 6 ft, performance dropped to 3/5 (0.74) as the wire approached the model's resolution limit. This 3–5 ft range provides roughly 2–3 steps of advance warning for a walking user.

C. Preliminary Walking Trial

Across fifteen walks, the system issued a pre-contact alert in 11 of 15 cases (73%). The 4 misses occurred on different participants and different walks, with little correlation to specific hazards. Survey results showed moderate Q1 scores (mean 3.4) indicating participants felt they generally had adequate reaction time, and moderate Q3 scores (mean 3.4) indicating the alerts were reasonably clear. Q2 scores were notably high (mean 4.4), reflecting strong agreement that the audio alert was disruptive. Participants also noted that a more descriptive alert such as indicating the type or proximity of the hazard would improve their ability to respond appropriately.

VI. DISCUSSION

A. Tile as a Preferred Deployment Surface

Tile outperformed both wood and carpet in raw detection confidence under bright conditions. Homes, kitchens, and bathrooms with tile flooring represent an ideal deployment environment for this system. The clean, uniform background maximizes the wire-to-floor contrast that the model relies on.

B. False Positives Are Surface and Lighting Dependent

False positives were concentrated on wood and tile in bright conditions, where the model misclassified linear features such as grout, lines, and cabinet edges as cables. This reflects a known limitation of lightweight models like YOLO11n when trained without sufficient hard-negative examples. Future datasets should explicitly include such negatives. Carpet produced no false positives, and all surfaces showed reduced errors in low light due to lower background contrast.

C. Effective Detection Range

The effective detection range is 3–5 ft. Below 1–2 ft the frame crops too tightly for reliable classification, and above 5–6 ft the wire resolves too narrowly for consistent detection.

D. Alert Fatigue is a Software-Fixable Design Flaw

The repeated audio alert firing is the single clearest finding from the user study and is entirely addressable in software. A debounce window of 2–3 seconds per detection event, or suppression until the bounding box leaves frame for at least one cycle, would directly reduce Q2 scores. This is limitation with a clear remediation path rather than a fundamental system problem.

E. Limitations

Testing was conducted in a single controlled residential environment. The participant sample ($n = 5$) is too small for statistical significance; results are preliminary. Tile was added as a surface after initial experimental design, and replication with a full balanced design across all three surfaces is needed. Dark condition performance, particularly for USB cables on wood, falls near the detection threshold and may not be reliable for real-world nighttime use without supplemental lighting.

REFERENCES

[1] K. L. Hsieh, L. Chen, and J. J. Sosnoff, ”Mobile Technology for Falls Prevention in Older Adults,” *The Journals of Gerontology: Series A*, vol. 78, no. 5, May 2022, doi: https://doi.org/10.1093/gerona/glac116.

[2] J. Pynoos, B. A. Steinman, and A. Q. D. Nguyen, “Environmental Assessment and Modification as Fall-Prevention Strategies for Older Adults,” textitClinics in Geriatric Medicine, vol. 26, no. 4, pp. 633–644, 2010, doi: https://doi.org/10.1016/j.cger.2010.07.001.

[3] M. Montero-Odasso, N. van der Velde, F. C. Martin, et al., ”World Guidelines for Falls Prevention and Management for Older Adults: A Global Initiative,” *Age and Ageing*, vol. 51, afac205, 2022, doi: https://doi.org/10.1093/ageing/afac205.

[4] C. Ohneberg, N. Stöbich, A. Warmbein, et al., ”Emerging Digital Technologies Used for Fall Detection in Older Adults in Aged Care: A Scoping Review,” *Journal of the American Medical Directors Association*, vol. 25, no. 11, pp. 105330, Nov. 2024, doi: https://doi.org/10.1016/j.jamda.2024.105330.

[5] A. García-Reyes, et al., ”A Comparative Overview of Technological Advances in Fall Detection Systems for Elderly People,” *MDPI Sensors*, vol. 25, no. 24, p. 7423, 2025, doi: https://doi.org/10.3390/s25247423.